Jobnity — Project Explanation Guide

Simple Project Walkthrough & Codebase Structure Reference

AI Career Platform | React 18 | Python FastAPI | NLP Taxonomy | Docker & Vercel

1. What is Jobnity? (Simple Explanation)

Jobnity is a smart career platform that helps job seekers find out exactly what skills they are missing for their dream job role, and gives them a clear, step-by-step learning roadmap to get hired.

> **The Problem It Solves**
>
> When students or job seekers upload their resume to traditional job sites, they only get generic search results. They don't know *why* they are missing out on interviews or *which specific technical skills* they need to learn to become job-ready.

> **The Jobnity Solution**
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> Jobnity reads the candidate's PDF or Word resume, automatically extracts their technical skills, compares them against **52 industry benchmark job roles** (like Data Scientist, Full Stack Developer, DevOps Engineer), calculates an accurate **Readiness Score**, and links them directly to courses for missing skills.

2. Every Process Explained Step-by-Step

1 Step 1: Resume Upload & Validation

The user selects their resume (PDF or DOCX format). The system checks that the file is valid, not corrupted, and under 10 MB in size. It saves the file safely in a temporary upload folder.

2 Step 2: Document Reading & Text Extraction

Python reads the file content line-by-line using two specialized tools:

- `pypdf` for PDF documents.
- `python-docx` for Microsoft Word documents (including tables and lists).

3 Step 3: NLP Skill & Contact Extraction

The system cleans the text (removing extra spaces/symbols) and scans it for skills. Instead of guessing, it matches words against a pre-built database of **1,156 technical skills** and **1,321 skill aliases** (e.g.

matching `JS` → `JavaScript`). It also finds the candidate's name, email, phone number, education, and work experience.

4 Step 4: User Profile Review & Skill Confirmation

Before running calculations, Jobnity presents the extracted profile to the candidate on screen. The candidate can add missing skills or remove incorrect ones. **The user's confirmed skill list becomes the single source of truth.**

5 Step 5: Weighted Skill Gap Analysis & Guardrails

The candidate selects a target job role (e.g., *DevOps Engineer*). The backend compares the user's confirmed skills against the role's required skills:

- **Guardrail Check:** Ensures distinct technologies are not confused (e.g., `Java` != `JavaScript`, `C` != `C++`, `Docker` != `Kubernetes`).
- **Weighted Scoring:** Required skills are weighted 1.5x more than optional preferred skills.
- **Output:** Calculates an exact **Readiness Score (%)** and **Skill Gap Score (%)**.

6 Step 6: Career Roadmap & Learning Resource Mapping

For every skill the candidate is missing, Jobnity queries its database of **125 curated learning resources** (Coursera, Udemy, YouTube tutorials, and official docs) and generates a structured step-by-step career path.

3. Codebase Structure — What Type of Code Do You Have?

Your codebase is organized into a clean, modern **Full-Stack Architecture**. Here is the breakdown:

Code Type	Technologies Used	Key Files & Directories	What It Does
Frontend (UI)	React 18, Vite, Framer Motion, Recharts, Vanilla CSS	`src/App.jsx` `src/pages/*.jsx` `src/components/*.jsx`	Renders the dark glassmorphic user interface, interactive skill gap charts, resume uploader, and mobile navigation.
Backend (Server)	Python 3, FastAPI, Uvicorn, Pydantic	`backend/app/main.py` `backend/app/api/routes/*.py`	Handles REST API requests, handles file uploads, runs text extraction, and processes skill matching algorithms.
NLP & Services	Regex, Python text processing	`backend/app/services/resume_parser.py` `backend/app/services/nlp_extractor.py` `backend/app/services/skill_gap.py`	Extracts text from PDF/DOCX, normalizes skill aliases, and calculates weighted readiness scores.
Database & Data	SQLite, SQLAlchemy ORM	`backend/jobnity.db` `jobnitydatasets/*.csv`	Stores 52 job roles, 1,156 skills, 1,321 aliases, and 125 learning resources. Stores candidate profiles and gap reports.
DevOps & Deploy	Docker, Vercel, Cloudflare Tunnel	`Dockerfile` `docker-compose.yml` `vercel.json` `api/index.py`	Packages the full app into Docker containers and allows serverless deployment on Vercel and secure HTTPS tunneling.

4. Quick Viva / Presentation Defense Guide

> **How to explain this project in 30 seconds:**
>
> *"Jobnity is a full-stack AI career platform built with React, FastAPI, and SQLite. It parses PDF and DOCX resumes, uses an NLP taxonomy normalizer with 1,320+ skill aliases to extract candidate profiles, and calculates weighted skill gap readiness scores against 52 benchmark job roles to generate personalized learning roadmaps."*